

Personas:

-Administrador

- Name: Carlos Nogueira
- Age: 42 years
- Sex:Male
- Profession:DataBase Engineer

-Description:

Carlos Nogueira is an experienced IT Administrator responsible for the maintenance, security, and efficiency of the Open Data Portal. With a background in database management and cybersecurity, he has worked in port administration IT for over a decade, ensuring that data systems remain robust, scalable, and well-protected. Carlos is deeply involved in database optimization, metadata tracking, and infrastructure management. He works behind the scenes to ensure that the system is always operational, efficiently handling large amounts of distributed and relational data. He is also responsible for monitoring system logs and troubleshooting errors.

-Goals and Needs:

Carlos Nogueira's primary goal is to maintain the security, reliability, and performance of the Open Data Portal. He needs to ensure that the system is always available, minimizing downtime and optimizing database performance. He must enforce user access control, ensuring that researchers, and data providers have the right permissions while preventing unauthorized access.

Another crucial need for Carlos is metadata tracking and provenance management. He is responsible for maintaining a system that records the history of each dataset, including its source, transformations, and updates. This allows users to trust the data and understand how it has changed over time. He also needs tools that allow users to inspect metadata efficiently, whether the data is stored in files, tables, or visualized in dashboards.

-Data Scientist(Logistics Analyst, Data consistency)

- Name:Mariana Costa
- Age:39
- Sex: Female
- Profession:Senior Data Scientist

-Description:

Mariana Costa is a Senior Data Scientist working as a consultant for both government agencies and private logistics companies. She has a Ph.D. in Maritime Studies and specializes in data-driven policy analysis and optimization of port operations. With 5 years of experience, she collaborates with port authorities and shipping companies to analyze efficiency, environmental impact, and economic trends in maritime logistics.

-Goals and Needs:

Mariana Costa needs structured, high-quality, and well-documented data. Access to metadata tracking is essential for her to understand data provenance and transformations. As a data scientist and policy advisor, she relies on advanced analytics and visualization tools. She wants to use the portal's dashboard to monitor key performance indicators like cargo throughput, vessel traffic, and environmental metrics. Ensuring data consistency and quality is another critical need for Mariana. She requires mechanisms to detect outliers and inconsistencies in datasets and tools that provide transparency on missing or estimated data. Being able to track the historical changes in datasets is essential for her work, as it allows her to assess the reliability of the information she analyzes.

Não sei se esta persona existe ou devia fazer parte do Administrador

-Data Provider (Port Authority Representative)?

-Name: João Silva

-Age: 32

-Sex: Male

-Profession:Port Worker

-Description:

João Silva is a Data Manager at the Port of Sines Administration, responsible for collecting, curating, and publishing maritime and logistics data on the Open Data Portal.

-Goals and Needs:

João's primary goal is to publish high-quality, well-documented, and reliable data on the Open Data Portal. He needs a system that allows him to efficiently upload and manage datasets while ensuring that metadata accurately describes data sources, transformations, and updates. João also needs tools for data quality control. He must be able to detect inconsistencies, missing values, and anomalies in the data before it is published. Automated outlier detection and error reporting are essential for preventing incorrect data from being shared with stakeholders.

Scenarios:

Scenario 1: Analyzing Port Congestion Trends for Policy Recommendations

Persona: Mariana Costa (Data Scientist)

Description:

Mariana has been tasked with advising a government agency on how to reduce congestion at a Port. She accesses the Open Data Portal to download historical data on vessel traffic, cargo processing times, and port delays. She identifies a pattern: congestion peaks correlate with inefficient cargo unloading times during specific shifts, so she prepares a report recommending optimized shift schedules and automation improvements.

Steps:

- 1- Mariana logs into the Open Data Portal.
- 2- Navigate through the dashboard to the page that contains the various datasets available.
- 3- Select the data for the port she wants.
- 4- Select which elements of the dataset she needs.
- 5- Filter the data to the desired timeframe.
- 6- Press the download Button and choose the format of the data.

Scenario 2: Assessing Data Quality and Provenance Before a Study

Persona: Mariana Costa (Data Scientist)

Description:

Before publishing a research paper on maritime supply chain disruptions, Mariana needs to verify the quality of the data she is using. She uses the Open Data Portal's metadata inspection tool to review the history of a dataset containing cargo arrival predictions. She notices that some data points were generated by an ML model rather than reported by human operators. To assess its reliability, she filters out ML-generated records and runs an independent statistical analysis then she documents her findings in the research paper.

Steps:

- 1- Mariana logs into the Open Data Portal.
- 2- Navigate through the dashboard to the page that does analytics of the datasets available.
- 3- Use the data provenance feature to identify which data-points are ML Generated.
- 4- Select the filter to not show the ML generated data-points.
- 5- Review the analyses provided.

Scenario 3: Ensuring Data Security and Quality in the Open Data Portal

Persona: Carlos Nogueira (DataBase Engineer)

Description:

Carlos Nogueira, the IT Administrator, receives a notification that a new dataset has been uploaded by a data provider. Before making it available to users, he must ensure that the data meets security, quality, and provenance standards. His goal is to validate the data's accuracy and check for inconsistencies and flag data points that have them .

Steps:

- 1- Carlos logs into the Open Data Portal's admin panel and sees a notification about a new dataset.
- 2- Opens the dataset and check its completeness.
- 3- Identifies that a data-point has a problem in one of the elements.
- 4- Uses the metadata inspection tool to verify the source of the data.
- 5- Flags the data to not be used in statistics until the problem is resolved.

Depois vai ser outro user storie a resolver este problema.

Também não sei se vamos ter isto necessariamente ou se só vai acontecer o cenário 4

Scenario 4: Uploading Data for Open Data Portal

Persona: João Silva (Data Provider)

Description:

João Silva, the Data Manager at the Port of Sines, has just collected new maritime cargo arrival data that needs to be uploaded to the Open Data Portal. He is responsible for ensuring that the data is accurate, complete, and properly documented. He passes the data through the ML and data provenance to properly mark the data with metadata.

Steps

- 1- João collects the latest maritime cargo data
- 2- Utilises a program to apply the ML model to complete empty data fields.
- 3- Passes the data through the metadata tool to document the origin of the data.

- 4- Accesses the Open Data Portal's.
- 5- Navigates to the upload data page.
- 6- Uploads the data.

Agora acontecia o cenário 3, mas também podia ser esta persona a fazer a verificação , mas se calhar tem mais lógica ser a outra.

User Stories:

Reunir as user stories das personas para fazer o diagrama alguns estão nos cenários.

Epic: DataScientist authentication

User Story 1: Create user account

As an Administrator,

I want to create accounts for users,

so that I can ensure only qualified personnel have access to the system.

Acceptance criteria:

- The administrator can access the User management section of the dashboard.
- The administrator can fill out a form to add information about the new user.
- The administrator can send an email through the system to the user.
- The DataScientist can activate their account.
- The Administrator can view the status of the user accounts.

Epic: User Interaction

User Story 2: Download Port data

As a Data Scientist,

i want to download Port DataSet,

so that i can use the data for a research paper

Acceptance criteria:

- The Data Scientist should be able login in the open data portal
- He should be able to navigate to the page of downloads
- He must be able to filter which part of the data set he wants to download
- The system must be able to respond with various formats
- The system should send a success message

Epic: User Interaction

User Story 3:Analyse Port data

As a Data Scientist,

i want to see various analysis of port activity,

so that i perform a logistical evaluation of the port.

Acceptance criteria:

- The Data Scientist should be able login in the open data portal
- He should be able to navigate to the page of analytics
- He must be able to filter which part of the data set he wants to analyze
- The system must be able to respond with graphs and tables to expose port data.

Epic: Managerial tasks

User Story 4: Manage user accounts

As a Administrator,
i want to see user activity,
so that I ensure system security.

Acceptance criteria:

- The administrator can access the User management section of the dashboard.
- The administrator should be able to see information about DataScientist and Data providers.
- He must be able to select a user.
- He must be able to see the user activity.
-

Epic: Managerial tasks

User Story 5: Flag bad data

As a Administrator,
i want to be able to flag data points,
so that it is not used in the analytic and downloads.

Acceptance criteria:

- The administrator can access the Data management section of the dashboard.
- The administrator should be able to see information about the data set.
- He must be able to filter by outliers .
- If the data seems incorrect he must be able to flag the data point.
- The flagged data point must not be used anywhere.

Epic: Managerial tasks

User Story 6: Remove Data

As a Administrator,
i want to remove incorrect data,
so that it is not used in the analytic and downloads.

Acceptance criteria:

- The admin must be able to communicate with the data provider to verify data.
- The administrator can access the Data management section of the dashboard.
- The administrator should be able to see information about the data set.
- He must be able to filter by flagged data points .
- He must be able to remove the data point.
-

Epic: Managerial tasks

User Story 7: Remove Flagged status

As a Administrator,
i want to remove the flagged status from the data point ,
so that the data can be used again in the data portal.

Acceptance criteria:

- The admin must be able to communicate with the data provider to verify data.
- The administrator can access the Data management section of the dashboard.
- The administrator should be able to see information about the data set.
- He must be able to filter by flagged data points .
- He must be able to remove the flagged status from the data point.

Epic: user interaction

User Story 8: Insert new data

As a Data Provider,
i want to add new data points into the dataset ,
so that the portal has more port information.

Acceptance criteria:

- The Data Provider should be able login in the open data portal
- He should be able to navigate to the page of post data.
- He must be able to insert the new data
- The system should send a success message.

Epic: user interaction

User Story 9: Analyse metadata

As a Data Scientist,
i want to be able to distinguish real data from ML generated data ,
so that I can evaluate the quality and accuracy of the ML model.

Acceptance criteria:

- The Data Scientist should be able login in the open data portal
- He should be able to navigate to the page of Data Provenance
- He must be able to filter out ML (and description) data origins.
- The system must be able to respond with graphs and tables to expose port data.

Epic: user interaction

User Story 10: Analyse outlier

As a Data Scientist,
i want to be able to visualise outlier data ,
so that I can evaluate if the data is correct.

Acceptance criteria:

- The Data Scientist should be able login in the open data portal
- He should be able to navigate to the page of analytics.
- The system must be able to respond with graphs and tables to expose port data.
- The graphs and tables must visually identify outlier data
- The user must be able to mouse over the data for more information

Uma outra podia ser tipo data report como fizemos no beachcontrol para avisar o manager sobre problemas

Non-functional requirements:

Performance: O sistema deve garantir tempos de resposta rápidos e uma execução eficiente das operações, mesmo sob alta carga de utilizadores e dados.

Security and Data Protection: O sistema deve proteger os dados contra acessos não autorizados, garantindo confidencialidade e integridade

Availability: O sistema deve estar disponível e operacional de forma contínua, minimizando o tempo de inatividade e garantindo alta confiabilidade.

Usability: O sistema deve proporcionar uma experiência intuitiva e acessível aos utilizadores, com uma interface amigável e um fluxo de interação eficiente.

Maintainability: O sistema deve ser fácil de atualizar, corrigir e expandir, garantindo um código modular e bem documentado.

Scalability: O sistema deve ser capaz de lidar com o crescimento da demanda, permitindo a adição de mais recursos sem perda significativa de desempenho.